

★ PRIMER · UNIT 2

Before the Lecture: How Data Becomes a Bar

Unit 2 · Financial Data Structure

Watch this first if OHLC and log returns are new

Twenty minutes of groundwork — what a bar is, why the clock is a poor ruler, and the one property every model quietly wishes returns had.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Bars and logs are defined here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 2 Sits

The course builds one pipeline; here is this unit's stone

Unit 1

done

What a model is, and why
finance is hard.

Unit 2

you are here

Rebuild the data itself — bars,
not the clock.

Unit 3

next

Fix the label — what are we
predicting?

The lecture cannot fix a label until the rows themselves are honest.

What a Bar Actually Is

A bar is a summary of many trades — four numbers standing in for a crowd.

The four numbers

- Open — the first trade.
- High and Low — the range.
- Close — the last trade.

Why summarise at all

- Raw ticks are millions of rows.
- Models need regular rows.
- A bar is one row.

Every dataset in this course is bars. The question is how you cut them.

The Words the Lecture Assumes

Five terms, and the lecture stops being foreign

Word	What it means	Example
Tick	one single trade	a buy at 10:03:22
Bar	a summary of many ticks	one trading day
OHLC	open, high, low, close	100 / 104 / 99 / 103
Volume	shares traded in the bar	1.2 million shares
Dollar volume	shares $\times$ price	\$124 million

Volume counts shares. Dollar volume counts money — and money is what moves.

Four Ways to Cut a Bar

The lecture compares all four; only one ignores the market

Bar type	A new bar starts when	On a quiet day you get
Time bar	the clock ticks over	just as many bars
Tick bar	N trades have happened	fewer bars
Volume bar	N shares have traded	fewer bars
Dollar bar	\$N has changed hands	fewer bars

Only the first ignores what the market is doing. That is the whole complaint.

Why the Clock Is a Poor Ruler

Markets do not arrive at a constant rate. A calendar samples a lull as heavily as a crash.

A quiet hour

- Almost nothing trades.
- A time bar fires anyway.
- You get a row of noise.

A crash

- A year of volume in minutes.
- One time bar swallows it.
- You lose the event.

Sample by activity and every row carries about the same amount of news.

Build Two Dollar Bars by Hand

A new bar every \$1,000 traded — keep the running total yourself

Trade	Price × Shares	Dollars	Running	Bar
t1	10.0×40	400	400	—
t2	10.5×40	420	820	—
t3	10.4×30	312	1,132	closes #1
t4	10.6×50	530	530	—
t5	10.8×45	486	1,016	closes #2

Two bars from five trades. On a quiet day the same five might be one bar.

Log Returns, and Why

The lecture uses logs throughout — this is the one line that explains it

Simple return

- $(\text{new} - \text{old}) / \text{old}$.
- +10% then -10% is not zero.
- They do not add up.

Log return

- The log of new over old.
- These do add across time.
- For small moves, nearly equal.

Log returns add; simple returns compound. That is the whole reason.

The Property Every Model Wishes For

The lecture calls it the IID dream — here is what the letters mean

Independent

- Today says nothing about tomorrow.
- Real returns nearly obey this.
- Nearly is doing work there.

Identically distributed

- The rules never change.
- Markets break this one badly.
- Regimes shift.

Good sampling gets you closer to the dream. Nothing gets you all the way.

How a Bar Becomes a Dataset

Three moves, in this order, before any model sees anything

1 · Sample

cut the bars

Time, tick, volume or dollar.
Dollar bars are the course
default.

2 · Clean

remove the lies

Bad prints, splits, and firms that
no longer exist.

3 · Transform

to returns

Prices become log returns, so
rows are comparable.

Skip step two and your model learns from a market that never existed.

Three Data Traps

All three are unit 2's own examples — recognise them before Wednesday

Trap	What happens	Why it survives
Survivorship	failed firms quietly dropped	returns look too good
Look-ahead	future information leaks in	accuracy looks too high
Bad tick	one impossible print stays	volatility looks wrong

Each one makes the backtest prettier. That is exactly why they last so long.

Notation You Will Meet

Three symbols carry the sampling section

Symbol	Read it as	It stands for
$\ln$	natural log	the log used for returns
Σ	sigma	add up what follows
r	r at t	the return of bar t

Sigma is only 'add these up'. It looks worse than it is.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>df.resample('1D')</code>	regroup rows into daily bars
<code>np.log(df['close'])</code>	take the natural log of a column
<code>.diff()</code>	subtract the previous row — log returns
<code>(px * vol).cumsum()</code>	running dollar volume, for dollar bars
<code>df.dropna()</code>	drop the empty first row

The whole bar pipeline is five lines. The judgement is in choosing them.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 3

1

"Time bars are the neutral choice"

They are the one choice that ignores the market. Neutral is not the same as default.

2

"Cleaning is a chore before the real work"

Cleaning decides what the model can possibly learn. In finance it is the real work.

3

"Log and simple returns are interchangeable"

They agree for small moves and diverge exactly when it matters — in a crash.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1 Five trades, a \$1,000 threshold. How many dollar bars?

two

2 Which bar type ignores what the market is doing?

time bar

3 Why do we take logs of prices?

returns add

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It rebuilds the row itself

Time, tick, volume and dollar bars — and what each one is blind to.

2

It adds an event filter

CUSUM — sample only when something has actually happened.

3

It ends in code

The problem set builds dollar bars and CUSUM events by hand, then in Python.

Next: Unit 2 lecture — Financial Data Structure